MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY 10545706

(PULSE WIDTH MULTIPLIER)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Pulse Width Multiplier Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation forbids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50785	Circuit Card Assemblies, General Specification for
MIL-c-50785/i	Circuit Card Assembly Voltage Comparator

DRAWINGS

U.S. Army, Frankford Arsenal

1054542S5	Sight Current Generator
10545455	Receptacle
10545481	Circuit Card-Assembly Voltage Comparator
10545706	Circuit Card Assembly Pulse Width Multiplier

PACKAGING DATA SHEET

10545706	Packaging of Circuit Card Assembly Pulse Width Multiplier

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1285

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3. REQUIREMENTS

3.1 Item definition. this assembly is a single ended printed circuit board consisting of a pulse width multiplier, a serrisoid generator and a reference voltage divider. The assembly is used to develop the sight correction portion of the torque current voltage when installed in the Sight Current Generator, Drawing Number 10545425.

3.1.1 Major component list The components of this assembly shall be as specified on the Engineering Parts List (EPL) 10545706.

3.1.2 Government furnished property list. Not applicable.

3.2 Characteristics

3.2,1 Performance. The assembly shall, when assembled in accordance with Drawing Number 1054S706, be capable of performing the following functions:

a. The pulse width multiplier, consisting of switching network Q1 through Q5, operational amplifiers Al and Az and a low pass filter network Ll, C12, shall form an analog signal proportional to the product of an input pulse train and an input dc voltage.

b. The serrosoid generator, consisting of operational amplifier A3 and q6 shall generate a serrosoid output .

c. The voltage reference divider shall provide outputs of +13 Vdc and -13 Vdc.

3.2.1.1 Input and output levels

3.2,1.1.1 Input power. The input power supplied shall be as specified in Table I under the conditions of Figure 1.

Table I. Input Voltage

Pin	Voltage
4 and 25	+15 ± 2.0 percent Vdc
14	-25 ± 2.0 percent Vdc
1 and 28	Ground

3.2.1.1.2 Output levels

3.2.1.1.2.1 Pin 2 output. The outputs between pin 2 and pin one shall be as specified in a through c. The maximum ripple voltage at pin 2, under any of the input conditions specified, shall be 50 millivolts peak-to-peak.

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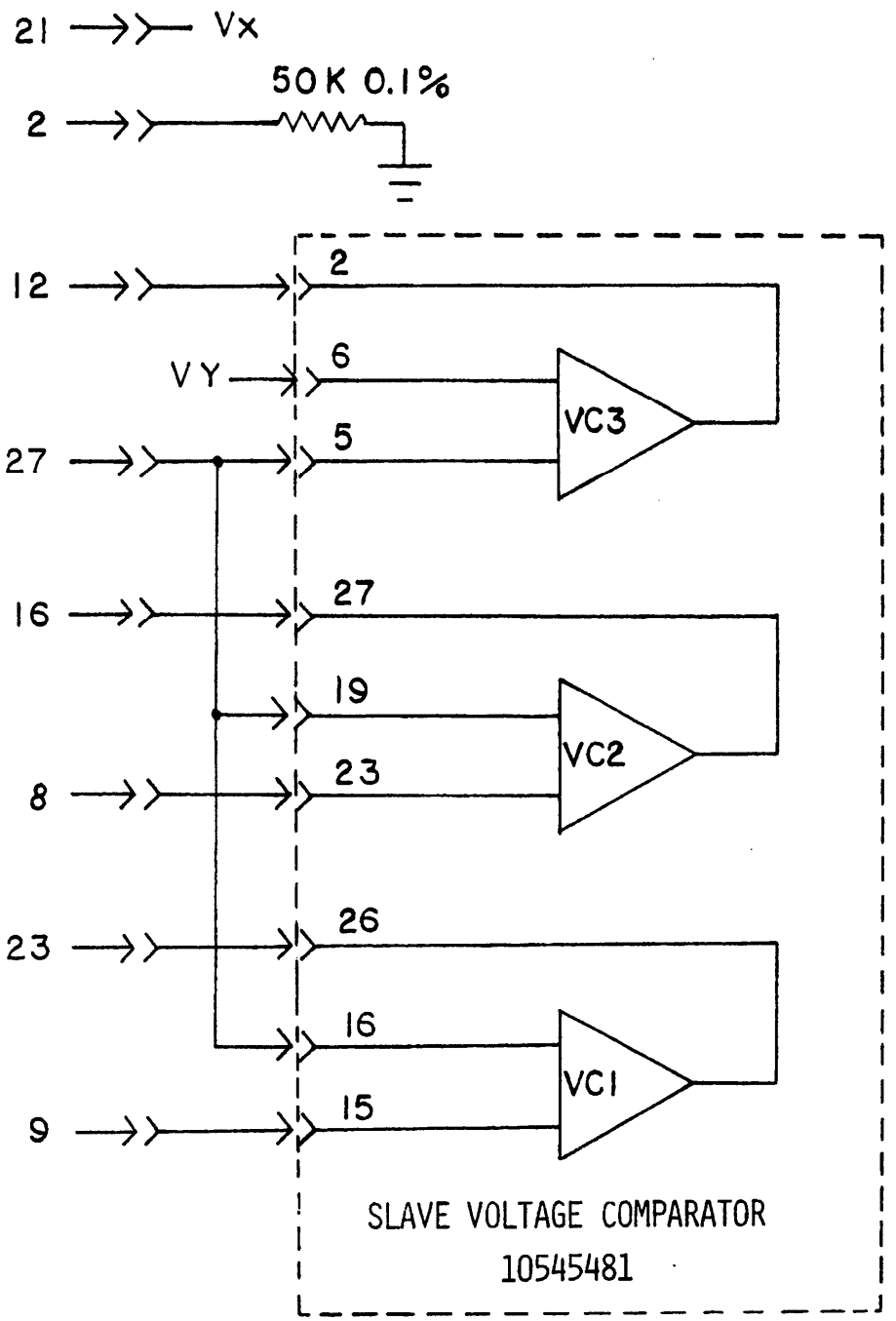


Figure 1. Input Connections

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8, Zero ± 0.020 Vdc when Vx input is zero Vdc and Vy input is between -13Vdc and +13Vdc,

b, Zero 2 0.200 Vdc when Vx input is zero to +13 Vdc and Vy input is zero Vdc,.

c. $-\left(\frac{Vx\ Vy}{13}\right)$ ± 2.0 percent when Vx input is between zero and +13 Vdc and Vy input is between -13 Vdc and +13 Vdc.

S,2.1,1,2.2 The output of pin 3 shall be satisfied when pin 2 meets its requirement.

3,2,1,1.2,3 Pin 8 output, The output shall be +13 Vdc ± 2.0 percent between pin 8 and pin one,

3.2.1,1,2.4 Pin 9 output, The output shall be -13 Vdc ± 2.0 percent between pin 9 and pin one,

3.2.1,1.2.S Pin 27 output. The output shall be a triangular wave with an amplitude from -13 ± 1.0 Vdc to +13 ± 1.0 Vdc within a period of 3.75 ± 1.25 milliseconds.

3.2.1.2 Environmental. l%e assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristic. The physical characteristics shall be as specified in drawing 10545706 and the engineering parts list.

3.3 Design and construction

3.3.1 Production drawings. This assembly shall be fabricated in accordance with the requirements of specification MIL-C-S0785, drawing 10545706 and" associated lists.

3.3.2 Standards of manufacture. The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship. The workmanship shall be as specified in specification MIL-C-5078S.

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4. QUALITY ASSURANCE PROVISION

4.1 General

4.1.1 Responsibility for inspection. The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination. The assembly shall undergo first article and control testing in accordance with 4,1.2 of specification MiL-C-50785.

4.2 Quality conformance inspections. The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785,

4.2.1 Test equipment

4.2.1.1 Power supply (PSI). The power supply shall be 15 k 0,1 Vdc, 100 milli-amperes or greater, ripple less than 10 millivolts (rev) rms or 100 mv peak-to-peak.

4.2,1.2 Power supply (PS2). The power supply shall be -15 ± 0.1 Vdc, 100 mini-amperes or greater, ripple less than 10 mv rms or 100 mv peak-to-peak.

4.2,1.3 Power supplies (PS, and PS~). The power supply shall be zero ± 13 Vdc, with accuracy of adjustment and regulation to ± .01 percent, 10 ma or greater.

4.2.1.4 Resistor. One resistor is required with a rating of 5000 ohms ± 1.0 percent, 1/4 watt.

4.2.2.5 Voltage comparator. One Circuit Card Assembly Voltage Comparator MIL-C-50785/l is required.

4.2,1.6 Digital voltmeter. The digital voltmeter (DVM) shall have a voltage accuracy and resolution of 0.01 percent or better from zero to 100 Vdc, Input impedance greater than 10 megohms on all voltage scales.

4.2.1.7 Oscilloscope. The oscilloscope shall have voltage, time accuracy and resolution of 5.0 percent or better, with a rise time of 100 nanoseconds or better and with external triggering. Voltage sensitivity shall include the range from 0.05 volts/cm to 5,0 volts/cm.

4.2.1.8 Receptacle. One Receptacle Drawing Number 10545455 is required.

4.2.2 Functional tests

4.2.2.1 Test connections. The power supplies shall be connected to the assembly under test in accordance with Figure 2 using receptacle 10545455. Power supply and test connections to the voltage comparator shall be as specified in MIL-C-50785/l drawing 10545481.

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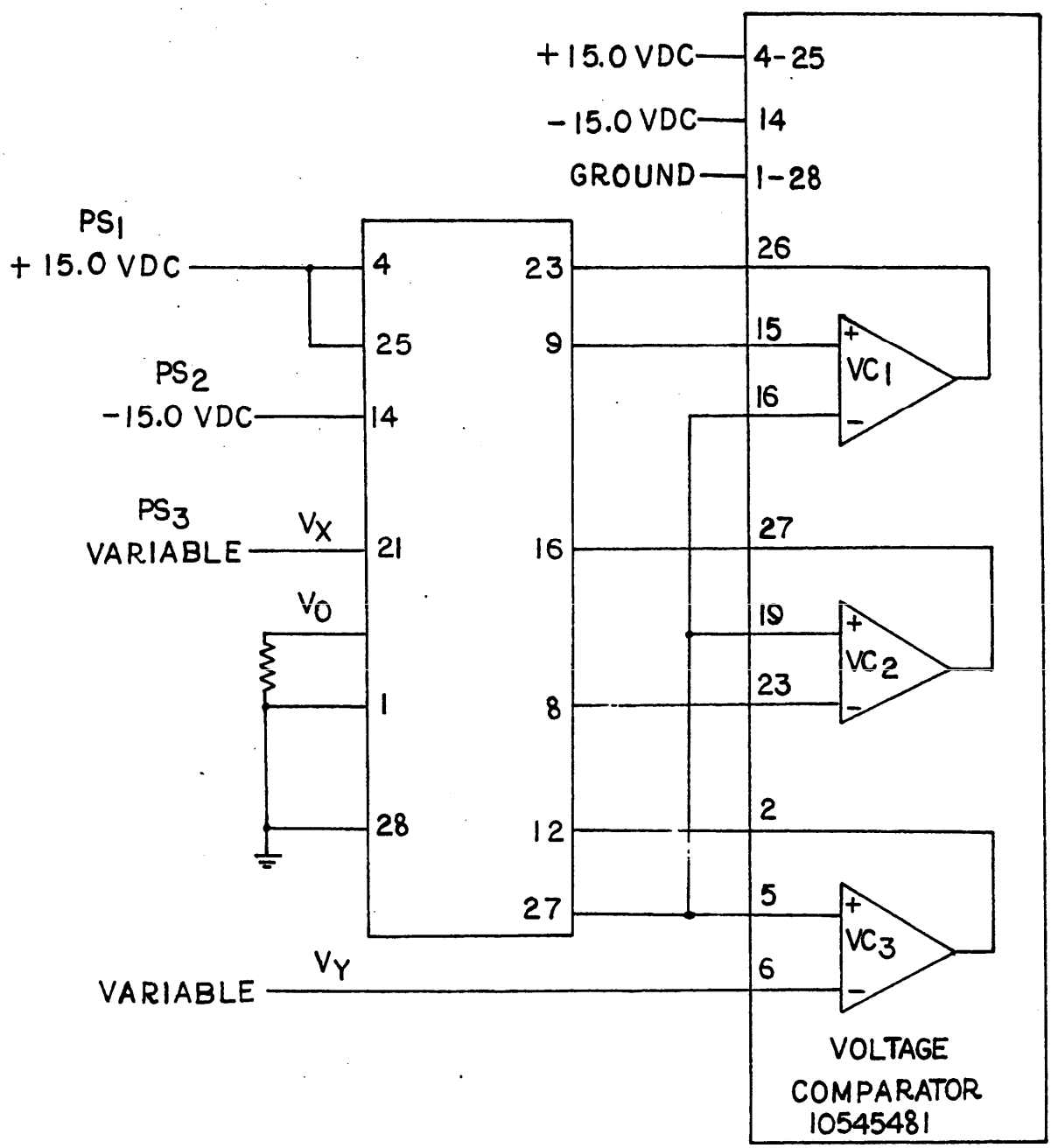


Figure 2. Test Connections

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4,2.2.2 Test procedure. Compliance with the requirements of 3.2.1,1 through 3.2.1.1.2.4 shall be verified as follows:

a. Adjust power supplies PS3 (Vx) and PS4 (Vy) as specified in Table 111. The voltage measured on pin 2 (Vo) shall be as specified. The ripple voltage on pin 2, as measured with the oscilloscope, shall be 50 millivolts peak-to-peak maximum for any of the input conditions specified,

b. The voltage on pin 8 shall be +13 Vdc ± 2.0 percent.

c, thee voltage on pin 9 shall be -13 Vdc ± 2.0 percent.

d. Using the oscilloscope verify the voltage waveform on pin 27 conforms to 3.2.1.1.2,4.

Table III. Test Limits

Input Vx (Vdc)	Input Vy (Vdc)	output Vo (Vdc)
+13. ± .013	+13. ± .013	-13. t .260
+13 ± .013	+10 ± .010	-lo ± ,200
+13 ± ,013	0 ± ,(-)10	o ± ,200
+13 ± 0013	.10 ± .010	+10 ± .200
+13 ± .013	.13 ± ,013	+13 ± .260
+10 ± .010 -	+13 ± .013	.10 ± .200
+10 ± .010	+10 ± .010	-7.692 ± .160
+10 ± .010	0 ± .010	() ± ,200
+10 ± ● 010	-lo ± .010	+7.692 ± .160
+10 ± .010	-13 ± .013	+10 ± .200
0 ± .010"	+13 ± .013	() ± ,020
0 ± .010	() ± .010"	0 ± .020
0 ± .010	-13 ± ● 013	o ± 020

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5. PREPARATION POR DELIVERY

5.1 Packaging. Packaging and marking shall be in accordance, with the level A requirements of Packaging Data Sheet (PDS), AMC **form 1029, 10545706.**

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable **to this specification.**

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Preparing Activity
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Project No. 1285-A316

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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